

COMS W4733: Computational Aspects of Robotics

Lecture 01 Notes: Rigid-Body Transformations

1 2D Translations and Rotations

1.1 Translations

- **Rigid-Body Transformation:** A mapping $h : \mathcal{W} \rightarrow \mathcal{W}$ (where $\mathcal{W} = \mathbb{R}^2$ or $\mathbb{R}^3$) that preserves relative distances and orientations (no mirror images) between all points on the body.

- **2D Translation Mapping:**

$$h(x, y, z) = (x + x_t, y + y_t, z + z_t)$$

- **Dual Viewpoint:** Translation of a body $\mathcal{A}$ by (x_t, y_t, z_t) is equivalent to an *inverse translation* of the world frame by (x_t, y_t, z_t) .

1.2 2D Rotations

- **Derivation:** Rotating a point (x, y) counterclockwise about the origin by angle θ yields (x', y') :

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \cos \theta - y \sin \theta \\ x \sin \theta + y \cos \theta \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = R(\theta) \begin{bmatrix} x \\ y \end{bmatrix}$$

- **Interpretation:** Columns of $R(\theta) = \begin{bmatrix} x_1 & y_1 \end{bmatrix}$ are the unit axes of the body frame expressed in the world frame.

- **Properties of 2D Rotation Matrices:**

1. **Orthogonality:** $R(\theta)^{-1} = R(\theta)^T = R(-\theta)$ (inverse rotation is a clockwise rotation by θ).
2. **Preserves Length/Area:** $|R(\theta)| = \det(R(\theta)) = +1$.
3. **Commutative:** $R(\theta_1 + \theta_2) = R(\theta_1)R(\theta_2) = R(\theta_2)R(\theta_1)$.

1.3 Example: 2D Relative Transformations

Consider three frames: World $\{s\}$, Body $\{b\}$, and Object $\{c\}$.

- Frame $\{b\}$ relative to $\{s\}$: Position $p = \begin{bmatrix} p_x \\ p_y \end{bmatrix}$, Rotation $P = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$
- Frame $\{c\}$ relative to $\{b\}$: Position $q = \begin{bmatrix} q_x \\ q_y \end{bmatrix}$, Rotation $Q = \begin{bmatrix} \cos \psi & -\sin \psi \\ \sin \psi & \cos \psi \end{bmatrix}$
- Frame $\{c\}$ relative to $\{s\}$:

$$r = p + Pq = \begin{bmatrix} p_x + q_x \cos \theta - q_y \sin \theta \\ p_y + q_x \sin \theta + q_y \cos \theta \end{bmatrix} \quad (\text{Affine mapping, not purely linear})$$

$$R = PQ = \begin{bmatrix} \cos(\theta + \psi) & -\sin(\theta + \psi) \\ \sin(\theta + \psi) & \cos(\theta + \psi) \end{bmatrix}$$

1.4 2D Homogeneous Transformations

To convert affine transformations $r = p + Rq$ into purely linear matrix multiplications, we embed points in **homogeneous coordinates** $(x, y, 1)^T$:

$$\begin{bmatrix} r_x \\ r_y \\ 1 \end{bmatrix} = \begin{bmatrix} p_x + q_x \cos \theta - q_y \sin \theta \\ p_y + q_x \sin \theta + q_y \cos \theta \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta & p_x \\ \sin \theta & \cos \theta & p_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} q_x \\ q_y \\ 1 \end{bmatrix} = T(\theta, p_x, p_y) \begin{bmatrix} q_x \\ q_y \\ 1 \end{bmatrix}$$

- T generalizes pure translations ($\theta = 0$) and pure rotations ($p = \mathbf{0}$).
- **Non-Commutativity:** Homogeneous matrices generally do not commute: $T_1 T_0 \neq T_0 T_1$.

2 3D Translations and Rotations

2.1 3D Principal Axis Rotations

Rotation in 3D is parameterized by three principal axis rotation matrices:

$$\begin{aligned} R_z(\alpha) &= \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} & (\text{about } z\text{-axis}) \\ R_y(\beta) &= \begin{bmatrix} \cos \beta & 0 & \sin \beta \\ 0 & 1 & 0 \\ -\sin \beta & 0 & \cos \beta \end{bmatrix} & (\text{about } y\text{-axis}) \\ R_x(\gamma) &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \gamma & -\sin \gamma \\ 0 & \sin \gamma & \cos \gamma \end{bmatrix} & (\text{about } x\text{-axis}) \end{aligned}$$

Properties: All 3D rotation matrices belong to $SO(3)$, satisfying $R^{-1} = R^T$ and $\det(R) = +1$.

2.2 Right-Hand Rule & Commutativity Comparison

- **Right-Hand Rule:** A positive angle θ corresponds to a counterclockwise rotation about the axis vector $\mathbf{v}$ (thumb points along $\mathbf{v}$, fingers curl in the positive rotation direction).
- **Commutativity Comparison:**
 - **2D Rotations:** Commutative ✓ $R(\theta)R(\phi) = R(\phi)R(\theta) = R(\theta + \phi)$
 - **3D Rotations:** Non-commutative × $R_y(\phi)R_z(\theta) \neq R_z(\theta)R_y(\phi)$

2.3 3D Homogeneous Transformation Matrix

A 3D rigid-body motion combining rotation $R \in SO(3)$ and translation $p \in \mathbb{R}^3$ is expressed as a 4×4 matrix acting on homogeneous coordinates $\mathbf{q} = (q_x, q_y, q_z, 1)^T$:

$$T = \begin{bmatrix} R_{3 \times 3} & p_{3 \times 1} \\ \mathbf{0}^T & 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & p_x \\ r_{21} & r_{22} & r_{23} & p_y \\ r_{31} & r_{32} & r_{33} & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Action Order: $T\mathbf{q} = R\mathbf{q} + p$ (Rotation first, then translation).

2.4 Composition Ordering Rules

When combining multiple transformations, the order of matrix multiplication depends on the reference frame:

Transformation Composition Rules

1. **Relative Transformations (Moving Frame):** If transform $A = T_j^i$ converts frame $i \rightarrow j$ (wrt frame i), and $B = T_k^j$ converts frame $j \rightarrow k$ (wrt frame j), then:

$$T_k^i = T_j^i T_k^j = AB$$

→ **Right-multiply (Post-multiply)** successive transformations specified relative to the current moving frame!

2. **Absolute Transformations (Fixed Frame):** If transformations A and B are specified relative to a single fixed reference frame (e.g., world frame):

$$C = BA$$

→ **Left-multiply (Pre-multiply)** successive transformations specified relative to a fixed frame!